

Operators as Dilators in the Proof-Theoretic Analysis of KP

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Table of Contents

1 Ordinal analysis

2 Dilators in proof theory

3 Operators as dilators

The realm of ordinal analysis

- Gentzen introduced ordinal analysis to prove the consistency of Peano arithmetic.
- The key tool is cut elimination: Eliminating Cuts from a given proof.

$$\frac{\Gamma, A \quad \Delta, \neg A}{\Gamma, \Delta} \text{Cut}$$

- Once we eliminate cuts, every formula occurring in the proof is built from formulas already occurring in the conclusion.

How to prove cut elimination?

We rank proofs by an ordinal α and apply induction on α :

$$\left| \frac{\alpha}{n+1} \right| \Gamma \implies \left| \frac{\omega^\alpha}{n} \right| \Gamma.$$

This gives

$$\left| \frac{\alpha}{n} \right| \Gamma \implies \left| \frac{\omega^{\overbrace{\omega \dots \omega}^{n \text{ times}} \alpha}}{0} \right| \Gamma.$$

Hence, every PA-proof has ordinal complexity $< \varepsilon_0$.

Kripke-Platek set theory KP

KP is a set theory of admissible sets, the basic universes of α -computability.

Definition

Kripke-Platek set theory KP is axiomatized by the following set of axioms:

- 1 Extensionality, Pairing, Union, Infinity.
- 2 Δ_0 -Separation: $\{x \in a \mid \phi(x, p)\}$ exists for a bounded ϕ .
- 3 Δ_0 -Collection: For a bounded formula ϕ ,

$$\forall a[\forall x \in a \exists y \phi(x, y, p) \implies \exists b \forall x \in a \exists y \in b \phi(x, y, p)].$$

- 4 $\in$ -induction scheme for every formula.

Collapsing

In the analysis of KP, a formal large ordinal Ω represents the 'height of the universe'.

Recall the statement of Δ_0 -collection: It pulls the behavior of the universe back to a small set:

$$\begin{array}{c} \forall x \in a \exists y \phi(x, y) \\ \Downarrow \\ \exists b \forall x \in a \exists y \in b \phi(x, y) \end{array}$$

Proof-theoretic collapsing has a similar flavor:

A derivation using the large ordinal Ω



A derivation bounded below Ω .

To perform the collapse, we must control which ordinal parameters occur in the derivation.

Operators

An operator is a closure operation

$$X \subseteq \text{Ord} \quad \longmapsto \quad \mathcal{H}(X) \subseteq \text{Ord}.$$

In an operator-controlled derivation

$$\mathcal{H} \Big|_{\rho}^{\alpha} \Gamma,$$

the operator controls the ordinal parameters available in the proof.

What does an operator do?

One obtains the impredicative cut-elimination:

$$\mathcal{H}_\eta \mid_{\Omega+1}^\alpha \Gamma \quad \Longrightarrow \quad \mathcal{H}_{\hat{\alpha}} \mid_{\psi_\Omega(\hat{\alpha})}^{\psi_\Omega(\hat{\alpha})} \Gamma.$$

- The large ordinal $\alpha \geq \Omega$ is collapsed into $\psi_\Omega(\hat{\alpha}) < \Omega$.
- The proof height is transformed.
- The operator becomes complicated, yet still keeps the relevant ordinal parameters available throughout the transformation.

Operators separate parameter control from the combinatorics of the proof.

What are dilators?

Roughly speaking, a dilator is a uniform ordinal notation system depending on a well-order X :

$$X \longmapsto D(X).$$

Its elements are terms

$$s(a), \quad a \in [X]^{<\omega},$$

where each term symbol s has a fixed finite arity.

The comparison

$$s(a) < t(b)$$

depends only on the relative order of the parameters in $a \cup b$, not on the particular well-order X .

Hence, for $f: X \rightarrow Y$,

$$D(X) \models s(a) < t(b) \implies D(Y) \models s(f[a]) < t(f[b]).$$

We also demand that $D(X)$ is a well-order when X is.

Examples of dilators

1 $\alpha \mapsto \alpha + \alpha, \alpha \mapsto \alpha^2.$

2 The epsilon flower

$$\alpha \mapsto \varepsilon(\alpha) := \varepsilon_\alpha.$$

3 The Feferman-Schütte flower

$$\alpha \mapsto \Gamma(\alpha) := \Gamma_\alpha,$$

where $\langle \Gamma_\alpha \mid \alpha \in \text{Ord} \rangle$ enumerates the common fixed points of the Veblen functions.

A dilator can be viewed as a uniform way of forming ordinals closed under certain operations.

Proofs varying with the ordinal parameter

Ordinary ordinal analysis assigns one ordinal to a proof:

$$\pi \longmapsto \alpha.$$

Girard instead considers proofs that vary uniformly with a well-order X :

$$X \longmapsto \pi(X).$$

The ordinal bound then also varies uniformly:

$$X \longmapsto D(X).$$

This is the basic idea behind β -proofs: one proof pattern, instantiated uniformly over different well-orders.

F -proofs

Girard also considered a notion of F -proof:

- β -proof is a functorial family of proofs

$$\langle \pi(\alpha) \mid \alpha \in \text{Ord} \rangle.$$

- F -proof is a functorial family of proofs

$$\langle \pi(F(\alpha)) \mid \alpha \in \text{Ord} \rangle.$$

F controls:

- 1 the ordinal domain over which each proof is formed;
- 2 the complexity of the family under proof transformations: cut elimination may replace F by a more complicated dilator F' .

Can operators be replaced by dilators?

Operators generate closures of ordinal parameters:

$$X \longmapsto \mathcal{H}(X).$$

Dilators also generate larger ordinal structures from smaller ones:

$$X \longmapsto D(X).$$

Both behave like uniform mechanisms for building ordinal structure.

Can operators in the analysis of KP be replaced by dilators?

Replace parameters by order templates

A fixed ordinal parameter loses information under reindexing.
Instead, replace actual parameters by a finite order template:

$$X = \{x_0 < x_1 < \cdots < x_{n-1}\}.$$

An embedding $f: X \rightarrow Y$ simply reinterprets the same template inside Y .

Thus the proof itself becomes functorial in the ordinal domain.

Dilator-controlled derivations

$$\mathcal{H} \left| \frac{\alpha}{\rho} \right. \Gamma \quad \rightsquigarrow \quad H[X] \left| \frac{\alpha}{\rho} \right. \Gamma$$

where

- H is a dilator; in the collapsing argument, H typically takes the form

$$H(X) = \varepsilon(X + F(X) + 1 + G(X)).$$

- $\alpha, \rho \in H(X)$.
- Γ is a sequent whose ordinal parameters occur in $H(X)$.

Functorial collapsing

The ordinary collapsing theorem becomes a functorial proof transformation:

$$H\text{-proof} \quad \Longrightarrow \quad H'\text{-proof},$$

where

$$H' = H \circ (\text{Id} + \psi_{\Omega} H (\text{Id} + \Omega)).$$

Thus impredicative cut elimination transforms not just an ordinal bound, but a dilator bound.

Application

Dilator-based analysis of KP provides an alternative proof of

Theorem (Girard)

Let $f: \omega_1^{\text{CK}} \rightarrow \omega_1^{\text{CK}}$ be a Σ_1 -over- $L_{\omega_1^{\text{CK}}}$ definable function. Then we can find a recursive dilator F such that

$$\forall \alpha < \omega_1^{\text{CK}} [f(\alpha) < F(\alpha)].$$

Questions?



Thank you!